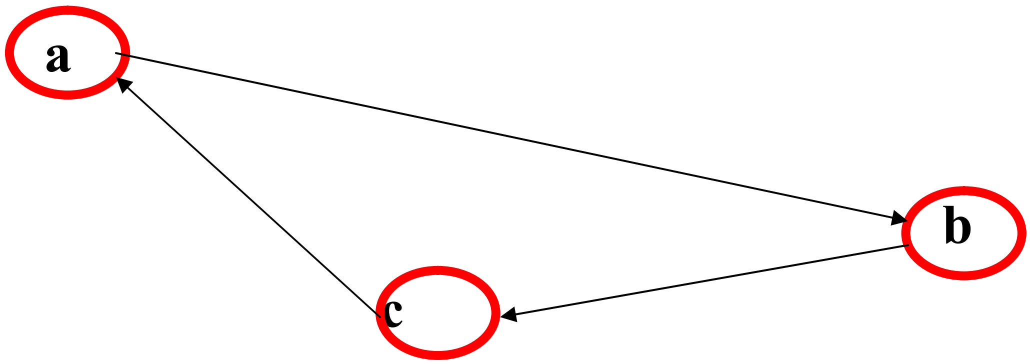

Graph (G^T)

Figure no 43

	a	b	c
a	0	0	1
b	1	0	0
c	0	1	0

Matrix of Graph G

	a	b	c
a	0	1	0
b	0	0	1
c	1	0	0

Matrix of Graph G^T

Matrix Decomposition Theorem

Matrix **decomposition** is a factorization of a matrix into some canonical form (Standard form/Normal form). There are many different matrix decompositions; each finds use among a particular class of problems.

Decompositions related to solving systems of linear equations

- LU decomposition
- Cholesky decomposition
- QR decomposition

Decompositions based on eigenvalues and related concepts

- **Eigen decomposition (also called spectral/diagonal decomposition)**
- Singular value decomposition

A. Eigen/diagonal Decomposition or spectral decomposition Theorem:

In linear algebra, eigendecomposition or sometimes spectral decomposition is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors. Only diagonalizable matrices can be factorized in this way.

Let A be the given Matrix, D is a diagonal matrix with the corresponding eigen values of Matrix A and **P be a matrix of eigenvectors** of a given Matrix A, Then A can be written as an eigen decomposition $\mathbf{A = PDP^{-1}}$

Example 67: Examine the eigen decomposition for the matrix $\mathbf{A} = \begin{pmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{pmatrix}$

Let $\mathbf{A} = \begin{pmatrix} \mathbf{4} & \mathbf{1} \\ \mathbf{1} & \mathbf{4} \end{pmatrix}$

$$\therefore \mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix}$$

$$|\mathbf{A} - \lambda\mathbf{I}| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1$$

$$\Rightarrow \quad |\mathbf{A} - \lambda\mathbf{I}| = 0$$

$$\Rightarrow \quad \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1 = 0$$

$$\Rightarrow (16 - 4\lambda - 4\lambda + \lambda^2) - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 16 - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\Rightarrow (\lambda - 5)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 5, \lambda = 3$$

$$\therefore \mathbf{D} = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Now to find out **the eigenvector** of the matrix A,

Let a non-zero vector v = $\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}$

We have, $\mathbf{Av} = \lambda\mathbf{v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = 3\begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 3]$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 4x+y &= 3x \\ x+4y &= 3y \end{aligned}$$

$$\Rightarrow \begin{aligned} x+y &= 0 \\ x+y &= 0 \end{aligned}$$

$$\Rightarrow x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 then x = -1

$\therefore$ Eigenvector $\mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for $\lambda = 3$

Again for $\lambda = 5$

Let a non-zero vector v = $\begin{bmatrix} x \\ y \end{bmatrix}$

We have, $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 5\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = 5\begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 5]$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix} = \begin{pmatrix} 5x \\ 5y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 4x+y &= 5x \\ x+4y &= 5y \end{aligned}$$

$$\Rightarrow \begin{aligned} -x+y &= 0 \\ x-y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x-y &= 0 \\ x-y &= 0 \end{aligned}$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular **(2-1) =1 free variable**, which is y.

Let y =1 , then x =1

$\therefore$ Eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda = 5$

P be a matrix of eigenvectors of a given Matrix A.

$$\therefore \mathbf{P} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

Now to find out $\mathbf{p}^{-1}$

$$\mathbf{c_{11} = 1,\ c_{12} = -1,\ c_{21} = -1,\ c_{22} = -1}$$

$$\mathbf{p}^c = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$\mathbf{Adj\,P} = (\mathbf{P}^C)^T = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$|\mathbf{P}| = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 = -2$$

$$\therefore \mathbf{P}^{-1} = \frac{\mathbf{Adj P}}{|\mathbf{P}|} = -\frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

Then A can be written as an eigen decomposition $\mathbf{A} = \mathbf{PDP}^{-1}$

Justification:

$$\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$= -\frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 3 & -3 \\ -5 & -5 \end{pmatrix}$$

$$= -\frac{1}{2}\begin{pmatrix} -3-5 & 3-5 \\ 3-5 & -3-5 \end{pmatrix}$$

$$= -\frac{1}{2}\begin{pmatrix} -8 & -2 \\ -2 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = \mathbf{A} \quad \text{(Proved)}$$

$\mathbf{A} = \mathbf{PDP}^{-1}$ (Proved)

Home task: Find eigen decomposition of the matrix $= \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$ and $= \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

Q-01: What is the difference between Eigen decomposition and SVD?

The SVD always exists for any sort of rectangular or square matrix, whereas the eigen decomposition can only exists for square matrices, and even among square matrices sometimes it doesn't exist.

<u>What is singular value of a Matrix?</u>

The singular values of a $\mathbf{m} \times \mathbf{n}$ matrix A are the square roots of the eigenvalues of the symmetric $\mathbf{n} \times \mathbf{n}$ matrix $\mathbf{A}^{\mathbf{T}}\mathbf{A}$. The Eigen values of $\mathbf{A}^{\mathbf{T}}\mathbf{A}$ are

$\lambda_1 \geq \lambda_2 \ldots\ldots\ldots\ldots\ldots \geq \lambda_r > \lambda_{r+1} \ldots\ldots\ldots\ldots\ldots\ldots \geq \lambda_n$.

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

$\sigma_i = \sqrt{\lambda_i}$ are called the singular value of A where i = 1, 2,……………, n.

It is customary to denote the singular values by $\sigma_{1,} \sigma_{2,} \sigma_{3,} \ldots\ldots\ldots\ldots \sigma_n$ and to list them in decreasing order:

$\sigma_{1,} \geq \sigma_2 \geq \sigma_3 \geq \ldots\ldots\ldots\ldots \geq \sigma_n$

Singular Value Decomposition:

In linear algebra, the **singular value decomposition** (**SVD**) is an important factorization of a rectangular real or complex matrix, with several applications **in signal processing , image processing and compression, gene expression analysis and statistics**. Applications which employ the SVD include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of eigenvectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

The singular value decomposition (SVD) is a generalization of the eigen-decomposition, which can be used to analyze rectangular matrices (the eigen- decomposition is defined only for squared matrices). By analogy with the eigen-decomposition, which decomposes a matrix into two simple matrices, the main idea of the SVD is to decompose a rectangular matrix into three simple matrices: Two orthogonal matrices and one diagonal matrix.

Matrix A is decomposed using SVD to acquire an orthogonal base in the multidimensional feature space:

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} * & \\ & * \\ & \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} * & * \\ * & * \end{bmatrix}}_{V^T}$$

$\mathbf{A = USV^T}$

Here,

$\mathbf{A \rightarrow}$ **Given Matrix**

$\mathbf{U \rightarrow}$ **Matrix of Eigen Vectors of the Matrix** $\mathbf{AA^T}$

$\mathbf{V \rightarrow}$ **Matrix of Eigen Vectors of the Matrix** $\mathbf{A^TA}$

$\mathbf{S \rightarrow}$ **Roots of Eigen Values of the Matrix** $\mathbf{AA^T}$ **or** $\mathbf{A^TA}$

$\mathbf{V^T \rightarrow}$ **Transpose of the Matrix** $\mathbf{V}$

Example 68: Compute Singular Value Decomposition (SVD) for the given matrix $\begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

Answer: Let, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

In order to find U, Now we need to find the eigenvectors for AA^T

Given, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\therefore \mathbf{A^T} = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

So, $\mathbf{AA^T} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 4+4 & 2-2 \\ 2-2 & 1+1 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \qquad \text{--------------------------------(i)}$$

Now we need to find the eigenvalues of $\mathbf{AA^T}$

$$\mathbf{AA^T - \lambda I} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{pmatrix}$$

$$\left|\mathbf{AA^T - \lambda I}\right| = 0$$

$$\Rightarrow \begin{vmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (8-\lambda)(2-\lambda) = 0$$

$$\Rightarrow (\lambda-8)(\lambda-2) = 0$$

$$\Rightarrow \lambda = 8,2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\boldsymbol{D} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $\boldsymbol{S} = \sqrt{\boldsymbol{D}} = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$

$$\mathbf{S} = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{---------------------------(ii)}$$

Now we need to find the eigenvector of $\mathbf{AA^T}$

We have, $\mathbf{Au_1} = \lambda\mathbf{u_1}$ $\quad\quad$ $[\boldsymbol{AV} = \lambda \boldsymbol{V}]$

Let $\mathbf{u_1} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

$\therefore \mathbf{AA^T u_1} = \lambda \mathbf{u_1}$

$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 8\begin{pmatrix} x \\ y \end{pmatrix}$ $\quad\quad$ [For $\lambda = \mathbf{8}$ & from (i) putting the value of $\boldsymbol{AA^T}$]

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 8x &= 8x \\ 2y &= 8y \end{aligned}$$

$$\Rightarrow \begin{aligned} 8x - 8x &= 0 \\ 2y - 8y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 8x &= 8x \\ -6y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 8x &= 8x \\ y &= 0 \end{aligned}$$

$\therefore \mathbf{y = 0}$

Let $\mathbf{x = -1}$

$\therefore \boldsymbol{u_1} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ It is a unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_1}| = \sqrt{(-1)^2 + (0)^2} = \sqrt{1} = 1$

Again, Let $\mathbf{u_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

Here,

$\therefore \mathbf{AA^T u_2} = \lambda \mathbf{u_2}$

$\begin{pmatrix} \mathbf{8} & \mathbf{0} \\ \mathbf{0} & \mathbf{2} \end{pmatrix}\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathbf{2}\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$ [For $\lambda = \mathbf{2}$ & from (i) putting the value of $\boldsymbol{AA^T}$]

$\Rightarrow \begin{pmatrix} \mathbf{8x} \\ \mathbf{2y} \end{pmatrix} = \begin{pmatrix} \mathbf{2x} \\ \mathbf{2y} \end{pmatrix}$

$\Rightarrow \mathbf{8x = 2x}$

$\mathbf{2y = 2y}$

$\Rightarrow \mathbf{8x - 2x = 0}$

$\mathbf{2y = 2y}$

$\Rightarrow \mathbf{6x = 0}$

$\mathbf{2y = 2y}$

$\therefore \boldsymbol{x} = \mathbf{0}$

$\mathbf{2}\boldsymbol{y} = \mathbf{2}\boldsymbol{y}$

Let $\mathbf{y = 1}$

$\therefore \boldsymbol{u_2} = \begin{pmatrix} \boldsymbol{x} \\ \boldsymbol{y} \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{1} \end{pmatrix}$ It is a unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_2}| = \sqrt{(\mathbf{0})^2 + (\mathbf{1})^2} = \sqrt{\mathbf{1}} = \mathbf{1}$

$\therefore \boldsymbol{U} = \begin{pmatrix} \boldsymbol{u_1} & \boldsymbol{u_2} \end{pmatrix} = \begin{pmatrix} \mathbf{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}$ ---------------------------------(iii)

Now we need to find the eigenvectors for $A^T A$

Given, $\mathbf{A} = \begin{pmatrix} \mathbf{2} & \mathbf{-2} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}$

$\therefore \mathbf{A^T} = \begin{pmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{-2} & \mathbf{1} \end{pmatrix}$

$\therefore \mathbf{A^T A} = \begin{pmatrix} \mathbf{2} & \mathbf{1} \\ \mathbf{-2} & \mathbf{1} \end{pmatrix}\begin{pmatrix} \mathbf{2} & \mathbf{-2} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}$

$= \begin{pmatrix} \mathbf{5} & \mathbf{-3} \\ \mathbf{-3} & \mathbf{5} \end{pmatrix}$ -----------------------------------(iv)

Now we need to find the eigenvalues of $A^T A$

$$\therefore A^T A - \lambda I = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{pmatrix}$$

$$\therefore \left| A^T A - \lambda I \right| = 0$$

$$\Rightarrow \begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)(5-\lambda) - 9 = 0$$
$$\Rightarrow 25 - 10\lambda + \lambda^2 - 9 = 0$$
$$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$$
$$\Rightarrow (\lambda - 8)(\lambda - 2) = 0$$
$$\Rightarrow \lambda = 8, 2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $S = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$

$$S = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{---------------------------(v)}$$

Now we need find the eigenvectors of $A^T A$ which are the columns of V.

We have, $Av_1 = \lambda v_1$ $\quad\quad\quad\quad$ $[AV = \lambda V]$

Let $v_1 = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\therefore A^T A v_1 = \lambda v_1$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 8 \begin{pmatrix} x \\ y \end{pmatrix}$$ [For $\lambda = 8$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\begin{pmatrix} 5x-3y \\ -3x+5y \end{pmatrix}=\begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 5x-3y=8x$$
$$-3x+5y=8y$$

$$\Rightarrow -3x-3y=0$$
$$-3x-3y=0$$

$$\Rightarrow x+y=0$$
$$x+y=0$$

$$\Rightarrow x+y=0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 $\therefore x=-1$

$\therefore v_1=\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} -1 \\ 1 \end{pmatrix}$; It is not unit vector or normal vector. Since the length of $\mathbf{v_1}$ is $|\mathbf{v_1}|=\sqrt{(-1)^2+(1)^2}=\sqrt{2}$; this vector $\mathbf{v_1}$ has to be converted into unit vector by dividing its magnitude or its length.

We have, $\mathbf{v_1}=\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$\therefore \mathbf{v_1}=\begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

$\mathbf{v_1}=\begin{pmatrix} \frac{-\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$ [*Multiplying by* $\sqrt{2}$ *on numerator and denominator]*

Similarly,

We have, $\mathbf{Av_2}=\lambda\mathbf{v_2}$ $[AV=\lambda V]$

Let $\mathbf{v_2}=\begin{pmatrix} x \\ y \end{pmatrix}$

$\therefore \mathbf{A^T Av_2}=\lambda\mathbf{v_2}$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 2\begin{pmatrix} x \\ y \end{pmatrix}$$ [For $\lambda = 2$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 3x - 3y &= 0 \\ -3x + 3y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x - y &= 0 \\ x - y &= 0 \end{aligned}$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 $\therefore x = 1$

$\therefore v_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of v_2 is $|v_2| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$; this vector v_2 has to be converted into unit vector by dividing its magnitude or its length.

We have,

$$v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Thus, $v_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

$\Rightarrow v_2 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$ *[Multiplying by $\sqrt{2}$ on numerator and denominator]*

Now we need to construct the augmented orthogonal matrix V,

$$V = \begin{pmatrix} v_1 & v_2 \end{pmatrix} = \begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\therefore V^T = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \qquad \text{------------------------------------(vi)}$$

From (ii), (iii) & (vi), Putting the values of U, S, V^T **in the relation A = USV^T**

Therefore **A = USV^T**

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \textit{Answer}$$

Justification Test:

A = USV^T

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{ [Putting the value of U }, V^T, \text{S]}$$

A = USV^T

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -2\sqrt{2}\times\frac{\sqrt{2}}{2}+0 & 2\sqrt{2}\times\frac{\sqrt{2}}{2}+0 \\ 0+\sqrt{2}\times\frac{\sqrt{2}}{2} & 0+\sqrt{2}\times\frac{\sqrt{2}}{2} \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} -1\times(-2)+0 & (-1)\times 2+0 \\ 0+1 & 0+1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \textit{ [Given Matrix] } \textbf{Proved}$$

Example 69: Compute Singular Value Decomposition (SVD) for the given matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find U, In order to find U, Now we need to find the eigenvectors for AA^T
Given,

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

So, $\mathbf{AA^T} = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}$ ------------------------------(i)

Now we need find the eigenvalues of $\mathbf{AA^T}$

$$\left|\mathbf{AA^T} - \lambda\mathbf{I}\right| = 0$$

$$\left|\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right| = 0$$

$$\left|\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right| = 0$$

Therefore,

$$\begin{vmatrix} 11-\lambda & 1-0 \\ 1-0 & 11-\lambda \end{vmatrix} = 0$$

$$(11-\lambda)(11-\lambda) - 1 = 0$$

$$\Rightarrow 121 - 11\lambda - 11\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 121 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 120 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda - 12\lambda + 120 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 12) = 0$$

$$\Rightarrow \lambda = 10, 12$$

Therefore eigenvalues are 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\boldsymbol{D} = \begin{pmatrix} 12 & 0 \\ 0 & 10 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $\boldsymbol{S} = \sqrt{\boldsymbol{D}} = \begin{pmatrix} \sqrt{12} & 0 \\ 0 & \sqrt{10} \end{pmatrix}$ ---------------------------(ii)

Now we need to find the eigenvector of AA^T

We have, $\mathbf{Au_1} = \lambda\mathbf{u_1}$ $\quad [AV = \lambda V]$

Let $\mathbf{u_1} = \begin{pmatrix} x \\ y \end{pmatrix}$

$\therefore \mathbf{AA^T u_1} = \lambda \mathbf{u_1}$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 12\begin{pmatrix} x \\ y \end{pmatrix}$ [For $\lambda = 12$ & from (i) putting the value of AA^T]

$$\begin{pmatrix} \mathbf{11} & \mathbf{1} \\ \mathbf{1} & \mathbf{11} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{12x} \\ \mathbf{12y} \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{11x+y} \\ \mathbf{x+11y} \end{pmatrix} = \begin{pmatrix} \mathbf{12x} \\ \mathbf{12y} \end{pmatrix}$$

$$\Rightarrow \mathbf{11x + y = 12x}$$
$$\mathbf{x + 11y = 12y}$$

$$\Rightarrow \mathbf{-x + y = 0}$$
$$\mathbf{x - y = 0}$$

$$\Rightarrow \mathbf{x - y = 0}$$
$$\mathbf{x - y = 0}$$

$$\Rightarrow \mathbf{x - y = 0}$$

Let y = 1 $\therefore \mathbf{x = 1}$

$\therefore \boldsymbol{u_1} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \mathbf{1} \\ \mathbf{1} \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_1}| = \sqrt{\mathbf{(1)^2 + (1)^2}} = \sqrt{\mathbf{2}}$; this vector $\mathbf{u_1}$ has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis, u_1 needs to be of length one. We divide $u_1$by its magnitude to accomplish this,

Thus, $\therefore \mathbf{u_1} = \begin{pmatrix} \frac{\mathbf{1}}{\sqrt{\mathbf{2}}} \\ \frac{\mathbf{1}}{\sqrt{\mathbf{2}}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{\mathbf{2}}$]

Again,

We have, $\boldsymbol{Au_2} = \lambda \boldsymbol{u_2}$ $\mathbf{[\mathit{AV} = \lambda \mathit{V}]}$

$\therefore \boldsymbol{AA^T u_2} = \lambda \boldsymbol{u_2}$

Let $\mathbf{u_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

For $\lambda = \mathbf{10}$

$$\begin{pmatrix} \mathbf{11} & \mathbf{1} \\ \mathbf{1} & \mathbf{11} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathbf{10} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$$

$\begin{pmatrix} \mathbf{11} & \mathbf{1} \\ \mathbf{1} & \mathbf{11} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathbf{10} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$ [For $\lambda = \mathbf{10}$ & from (i) putting the value of $\boldsymbol{AA^T}$]

$$\begin{pmatrix} \mathbf{11} & \mathbf{1} \\ \mathbf{1} & \mathbf{11} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \begin{pmatrix} \mathbf{10x} \\ \mathbf{10y} \end{pmatrix}$$

$$\begin{pmatrix} 11x+y \\ x+11y \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \end{pmatrix}$$

$$\Rightarrow \begin{aligned} 11x+y &= 10x \\ x+11y &= 10y \end{aligned}$$

$$\Rightarrow \begin{aligned} x+y &= 0 \\ x+y &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x+y &= 0 \\ x+y &= 0 \end{aligned}$$

$$\Rightarrow x+y=0$$

Let y = -1 $\therefore \mathbf{x = 1}$

$\therefore \boldsymbol{u_2} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{u_2}$ is $|\mathbf{u_2}| = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$; this vector $\mathbf{u_2}$ has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis, u_2 needs to be of length one. We divide u_2 by its magnitude to accomplish this,

Thus, $\therefore \mathbf{u_2} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

Since $\mathbf{u_2}$ is already of length one, dividing by the magnitude we get the same vector back,

Setting up the augmented matrix U

$$\mathbf{U} = (\mathbf{u_1} \quad \mathbf{u_2}) = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \qquad \text{------------------------------------(iii)}$$

Now we need to find the eigenvectors of $A^T A$ which are the columns of V.

Start with the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find V, we have to start with $\mathbf{A^T A}$ the transpose of A is

$$\mathbf{A^T} = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$\therefore \mathbf{A^TA} = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$= \begin{pmatrix} 3\times3+1 & 3-3 & 3-1 \\ 3-3 & 1+9 & 1+3 \\ 3-1 & 1+3 & 1+1 \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \qquad \text{---(iv)}$$

Now we need to find the eigenvalue of $\mathbf{A^TA}$

$$\left|\mathbf{A^TA} - \lambda \mathbf{I}\right| = 0$$

$$\left|\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right| = 0$$

$$\left|\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}\right| = 0$$

Therefore,

$$\left|\begin{pmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{pmatrix}\right| = 0$$

$$\begin{vmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{vmatrix} = 0$$

$$(10-\lambda)\{(10-\lambda)(2-\lambda)-16\} + 2\{0-2(10-\lambda)\} = 0$$

$$\Rightarrow (10-\lambda)(20-10\lambda-2\lambda+\lambda^2-16) - 4(10-\lambda) = 0$$

$$\Rightarrow (10-\lambda)(20-10\lambda-2\lambda+\lambda^2-16-4) = 0$$

$$\Rightarrow (10-\lambda)(\lambda^2-12\lambda) = 0$$

$$\Rightarrow \lambda(10-\lambda)(\lambda-12) = 0$$

$$\Rightarrow \lambda = 0,10,12$$

Therefore eigenvalues are 0, 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\boldsymbol{D} = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives, The matrix of singular values $S = \sqrt{D} = \begin{pmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{pmatrix}$ -------------------------------(v)

Now we need to find the eigenvector of A^TA which, are the columns of V.

We have, $Av_1 = \lambda v_1$ $\quad\quad\quad\quad$ $[AV = \lambda V]$

$\therefore A^T A v_1 = \lambda v_1$

Let $\mathbf{v_1} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$

$$\mathbf{A^T A v_1 = \lambda v_1}$$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 12 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{[For } \lambda = \mathbf{12} \text{ \& from (iv) putting the value of } A^T A\text{]}$$

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \\ 12z \end{pmatrix}$$

$$\Rightarrow 10x + 2z = 12x$$
$$10y + 4z = 12y$$
$$2x + 4y + 2z = 12z$$

$$\Rightarrow 2x + 4y + 2z = 12z$$
$$10x + 2z = 12x$$
$$10y + 4z = 12y$$

$$\Rightarrow 2x + 4y - 10z = 0$$
$$-2x + 2z = 0$$
$$-2y + 4z = 0$$

$$\Rightarrow 2x + 4y - 10z = 0$$
$$-2x + 0.y + 2z = 0$$
$$-2y + 4z = 0$$

We have,

$\mathbf{L_i \to -a_{i1}L_1 + a_{11}L_i}$

i = 2,

$\mathbf{L_2 \to -a_{21}L_1 + a_{11}L_2}$

$\to$ -(-2)(2x + 4y-10z)+2(-2x + 0.y + 2z)

$\to$ 4x + 8y - 20z - 4x + 0 + 4z

$\to$ 8y -16z

Thus we obtain a new system,

$$\begin{aligned} 2x + 4y - 10z &= 0 \\ 8y - 16z &= 0 \\ -2y + 4z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 2x + 4y - 10z &= 0 \\ y - 2z &= 0 \\ y - 2z &= 0 \end{aligned}$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$\begin{aligned} 2x + 4y - 10z &= 0 \\ y - 2z &= 0 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2) =1 free variable, which is z.

Here free variable is z, Let z = 1, $\therefore$ **y = 2, then** $\mathbf{2x + 4 \times 2 - 10 \times 1 = 0}$

$$\therefore \mathbf{x = 1}$$

$$\therefore v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$|\mathbf{v_1}| = \sqrt{(1)^2 + (2)^2 + (1)^2} = \sqrt{6}$$

Since V has an orthonormal basis, v_1 needs to be of length one. We divide v_1 by it's magnitude to accomplish this, Thus, $\therefore \mathbf{v_1} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$

Again,

We have, $Av_2 = \lambda v_2$ $\quad\quad\quad\quad$ $[AV = \lambda V]$

$\therefore A^T A v_2 = \lambda v_2$

Let $\mathbf{v_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$

$\mathbf{A^T A v_2 = \lambda v_2}$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 10\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$ [For $\lambda = \mathbf{10}$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \\ 10z \end{pmatrix}$$

$$\begin{aligned} \Rightarrow\ & 10x + 2z = 10x \\ & 10y + 4z = 10y \\ & 2x + 4y + 2z = 10z \end{aligned}$$

$$\begin{aligned} \Rightarrow\ & 2x + 4y + 2z = 10z \\ & 10x + 2z = 10x \\ & 10y + 4z = 10y \end{aligned}$$

$$\begin{aligned} \Rightarrow\ & 2x + 4y - 8z = 0 \\ & 2z = 0 \\ & 4z = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow\ & 2x + 4y - 8z = 0 \\ & 2z = 0 \\ & 2z = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow\ & 2x + 4y - 8z = 0 \\ & 2z = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow\ & 2x + 4y - 8z = 0 \\ & z = 0 \end{aligned}$$

$$\therefore z = 0$$

Then $\Rightarrow 2x + 4y - 8 \times 0 = 0$

$$\Rightarrow 2x + 4y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Here free variable is y, Let y = -1, then x = 2

$$\therefore v_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$$

$$|v_2| = \sqrt{(2)^2 + (-1)^2 + (0)^2} = \sqrt{5}$$

Since V has an orthonormal basis, ν_2 needs to be of length one. We divide ν_2 by it's magnitude to accomplish this, Thus, $\therefore \mathbf{v_2} = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} \\ 0 \end{pmatrix}$

Again,

We have, $\boldsymbol{Av_3 = \lambda v_3}$ $\qquad\qquad$ $[\boldsymbol{AV = \lambda V}]$

We have, $\mathbf{A^T A v_3 = \lambda v_3}$

Let $\mathbf{v_3} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$

$$\mathbf{A^T A v_3 = \lambda v_3}$$

$$\begin{pmatrix} \mathbf{10} & \mathbf{0} & \mathbf{2} \\ \mathbf{0} & \mathbf{10} & \mathbf{4} \\ \mathbf{2} & \mathbf{4} & \mathbf{2} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix} = 0 \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \\ \mathbf{z} \end{pmatrix}$$ [For $\lambda = \mathbf{0}$ & from (iv) putting the value of $\boldsymbol{A^T A}$]

$$\begin{pmatrix} \mathbf{10x + 0 + 2z} \\ \mathbf{0 + 10y + 4z} \\ \mathbf{2x + 4y + 2z} \end{pmatrix} = \begin{pmatrix} \mathbf{0x} \\ \mathbf{0y} \\ \mathbf{0z} \end{pmatrix}$$

$$\Rightarrow \mathbf{10x + 2z = 0}$$
$$\mathbf{10y + 4z = 0}$$
$$\mathbf{2x + 4y + 2z = 0}$$
$$\Rightarrow \mathbf{2x + 4y + 2z = 0}$$
$$\mathbf{10x + 2z = 0}$$
$$\mathbf{10y + 4z = 0}$$
$$\Rightarrow \mathbf{2x + 4y + 2z = 0}$$
$$\mathbf{10x + 0.y + 2z = 0}$$
$$\mathbf{10y + 4z = 0}$$

$$\mathbf{L_i \to -a_{i1}L_1 + a_{11}L_i}$$

i = 2,

$$\mathbf{L_2 \to -a_{21}L_1 + a_{11}L_2}$$
$$\to \text{-10(2x + 4y+2z)+2(10x + 0.y + 2z)}$$
$$\to \text{-20x - 40y - 20z +20x + 0 + 4z}$$
$$\to \text{-40y -16z}$$

Thus we obtain the following system:

$$\Rightarrow \begin{aligned} 2x + 4y + 2z &= 0 \\ -40y - 16z &= 0 \\ 10y + 4z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 2x + 4y + 2z &= 0 \\ 10y + 4z &= 0 \\ 10y + 4z &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} 2x + 4y + 2z &= 0 \\ 10y + 4z &= 0 \end{aligned}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2) =1 free variable, which is z

Here free variable is z, Let z = -5, then y = 2, then x =1

$$\therefore v_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -5 \end{pmatrix}$$

$$|v_3| = \sqrt{(1)^2 + (2)^2 + (-5)^2} = \sqrt{30}$$

Since V has an orthonormal basis, v_3 needs to be of length one. We divide v_3 by it's magnitude to accomplish this, Thus, $\therefore v_3 = \begin{pmatrix} \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} \\ \frac{-5}{\sqrt{30}} \end{pmatrix}$

Now we need to construct the augmented orthogonal matrix V,

$$V = (v_1 \quad v_2 \quad v_3)$$

All this to give us

$$V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

when we really want its transpose

$$V^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

For S we take the square roots of the non-zero eigenvalues and populate the diagonal with them, putting the largest in s_{11}, the next largest in s_{22} and so on until the smallest value ends up in s_{mn}. The non-zero eigenvalues of U and V are always the same, so that's why it doesn't matter which one we take them from. Because we are doing full SVD, instead of reduced SVD (next section), we have to add a zero column vector to S so that it is of the proper dimensions to allow multiplication between U and V. The diagonal entries in S are the singular values of A, the columns in U are called left singular vectors, and the columns in V are called right singular vectors.

$$S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix}$ [Ignoring third row of the matrix as all elements are zero in the last row]

Now we have all the pieces of the puzzle

$A = USV^T$

Putting the value of U , V^T , S in $A = USV^T$

$A = USV^T$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{12}}{\sqrt{2}} & \frac{\sqrt{10}}{\sqrt{2}} & 0 \\ \frac{\sqrt{12}}{\sqrt{2}} & \frac{-\sqrt{10}}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} \mathbf{3} & \mathbf{1} & \mathbf{1} \\ \mathbf{-1} & \mathbf{3} & \mathbf{1} \end{bmatrix} \text{[Proved]}$$

.

Home Task:
Find the singular value decomposition of the matrix

i. $\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{0} \end{bmatrix}$ ii. $\boldsymbol{A} = \begin{bmatrix} \mathbf{2} & \mathbf{1} & \mathbf{-2} \end{bmatrix}$

iii. $\boldsymbol{A} = \begin{bmatrix} \mathbf{2} & \mathbf{4} \\ \mathbf{1} & \mathbf{3} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$

Hints:

$$U = \begin{bmatrix} 0.82 & -0.58 & 0 & 0 \\ 0.58 & 0.82 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.40 & -0.91 \\ 0.91 & 0.40 \end{bmatrix}$$

$$S = \begin{bmatrix} 5.47 & 0 \\ 0 & 0.37 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Singular Value Decomposition:

The Singular Value Decomposition is an important factorization of a real or complex matrix, with several applications in signal processing and statistics. Applications, which employ the SVD, include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of Eigen vectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

Algorithm of Singular Value Decomposition:
To find the Singular Value Decomposition of the matrix:

1. Find the eigenvalues of the matrix $\mathbf{A^T A}$ and arrange them in descending order
2. Find the number of nonzero eigenvalues of the matrix $\mathbf{A^T A}$
3. Find the orthogonal eigenvectors of the matrix $\mathbf{A^T A}$ corresponding to the obtained eigenvalues, and arrange them in the same order to form the column-vectors of the matrix V
4. Form a diagonal matrix $\mathbf{S} = \sum = \sigma$ placing on the leading diagonal of it the square roots $\sigma_i = \sqrt{\lambda_i}$ of $\mathbf{p = min\{m,n\}}$ first eigenvalues of the matrix $\mathbf{A^T A}$ got in I in descending order
5. Find the first column-vectors of the matrix U, $\mathbf{u_i = \sigma_i^{-1} A v_i (i = 1 : r)}$
6. Add to the matrix U the rest of m-r vectors using the Gram-Schmidt Orthogonalization process

Image Processing with SVD

We want to approximate the $\mathbf{m \times n}$ matrix A by using far fewer entries than in the original matrix.

$$\mathbf{A = USV^T = SUV^T}$$
$$\Rightarrow \mathbf{A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \ldots\ldots\ldots\ldots\ldots\ldots\ldots + \sigma_r u_r v_r^T + 0.u_{r+1} v_{r+1}^T + \ldots\ldots\ldots}$$

Since the singular values are always greater than zero. Adding on the dependent terms where the singular values are equal to zero does not effect the image. The terms at the end of the equation zero out leaving us with:

$$\Rightarrow \mathbf{A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \ldots\ldots\ldots\ldots\ldots\ldots\ldots + \sigma_r u_r v_r^T}$$

Conclusion: We can further approximate the matrix (original image) by leaving off more singular terms of the matrix A. Since the singular values are arranged in descending order, the last terms will have the least effect on the overall image.

Doing this (leaving more singular values) reduces the amount of space required to store the image on a computer.

Justification using above example:

The original image is: $\mathbf{A} = \begin{pmatrix} \mathbf{2} & \mathbf{-2} \\ \mathbf{1} & \mathbf{1} \end{pmatrix}$

$\mathbf{S} = \begin{pmatrix} \mathbf{2\sqrt{2}} & \mathbf{0} \\ \mathbf{0} & \mathbf{\sqrt{2}} \end{pmatrix}$ **i.e.** $\mathbf{\sigma_1 = 2\sqrt{2}, \quad \sigma_2 = \sqrt{2}}$

$U = (u_1 \;\; u_2) = \begin{pmatrix} \mathbf{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \end{pmatrix}$

$V = (v_1 \;\; v_2) = \begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T$$
$$= 2\sqrt{2}\begin{pmatrix} -1 \\ 0 \end{pmatrix}\begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \sqrt{2}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$
$$= \begin{pmatrix} -2\sqrt{2} \\ 0 \end{pmatrix}\begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix}\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$
$$= \begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} = A \text{ (Original Image can be obtained)}$$

We can get approximate image by using less singular values. Say number of singular values are 10 i.e. $\sigma_1, \sigma_2, \ldots\ldots\ldots\ldots\ldots\ldots\ldots\sigma_{10}$.

Then,

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \ldots\ldots\ldots\ldots\ldots\ldots + \sigma_{10} u_{10} v_{10}^T$$

If we use 5 singular values then

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \ldots\ldots\ldots\ldots\ldots\ldots + \sigma_5 u_5 v_5^T$$

In this result we can get apprximate image of the original image (A)

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

What is the use of SVD?

01. SVD can be used to compute optimal low-rank approximations of arbitrary matrices.
02. Face recognition: represent the face images as eigenfaces and compute distance between the query face image in the principal component space.

B. LU Decomposition

A procedure for decomposing a $N \times N$ matrix A into a product of a lower triangular matrix L and an upper triangular matrix U,

$$A = LU$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}\begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

LU decomposition is implemented in *Mathematica* as LU Decomposition[*m*].

Example 70

Solve the linear system of equations using LU decomposition

$$u_1 + 2u_2 + 3u_3 = 5$$
$$2u_1 - 4u_2 + 6u_3 = 18$$
$$3u_1 - 9u_2 - 3u_3 = 6$$

Solution:

Given the linear system of equations

$$u_1 + 2u_2 + 3u_3 = 5$$
$$2u_1 - 4u_2 + 6u_3 = 18$$
$$3u_1 - 9u_2 - 3u_3 = 6$$

This can be written as

$\mathbf{Au} = \mathbf{b}$ ------------------------------(i)

Where,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \mathbf{u}_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

Again, According to LU decomposition,

$\mathbf{A} = \mathbf{LU}$ ------------------------------(ii)

Where,

$$\mathbf{L} = \begin{pmatrix} \mathbf{l}_{11} & 0 & 0 \\ \mathbf{l}_{21} & \mathbf{l}_{22} & 0 \\ \mathbf{l}_{31} & \mathbf{l}_{32} & \mathbf{l}_{33} \end{pmatrix} \quad \text{and} \quad \mathbf{U} = \begin{pmatrix} 1 & \mathbf{u}_{12} & \mathbf{u}_{13} \\ 0 & 1 & \mathbf{u}_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

From (ii), $\mathbf{A} = \mathbf{LU}$

$\Rightarrow \mathbf{LU} = \mathbf{A}$

$$\begin{pmatrix} \mathbf{l}_{11} & 0 & 0 \\ \mathbf{l}_{21} & \mathbf{l}_{22} & 0 \\ \mathbf{l}_{31} & \mathbf{l}_{32} & \mathbf{l}_{33} \end{pmatrix} \begin{pmatrix} 1 & \mathbf{u}_{12} & \mathbf{u}_{13} \\ 0 & 1 & \mathbf{u}_{23} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\begin{pmatrix} l_{11}+0+0 & l_{11}u_{12}+0+0 & l_{11}u_{13}+0+0 \\ l_{21}+0+0 & l_{21}u_{12}+l_{22}+0 & l_{21}u_{13}+l_{22}u_{23}+0 \\ l_{31}+0+0 & l_{31}u_{12}+l_{32}+0 & l_{31}u_{13}+l_{32}u_{23}+l_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

Now,

We can write

$l_{11} = 1$ -------------------------------------(iii)

$l_{11}u_{12} = 2$ -------------------------------------(iv)

$l_{11}u_{13} = 3$ -------------------------------------(v)

$l_{21} = 2$ -------------------------------------(vi)

$l_{21}u_{12} + l_{22} = -4$ -------------------------------------(vii)

$l_{21}u_{13} + l_{22}u_{23} = 6$ -------------------------------------(viii)

$l_{31} = 3$ -------------------------------------(ix)

$l_{31}u_{12} + l_{32} = -9$ -------------------------------------(x)

$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3$ -------------------------------------(xi)

From (iii) & (iv)

$l_{11} = 1$

$l_{11}u_{12} = 2$

$1.u_{12} = 2$

$u_{12} = 2$ -------------------------------------(xii)

From (iii) & (v)

$l_{11}u_{13} = 3$

$1.u_{13} = 3$

$u_{13} = 3$ -------------------------------------(xiii)

From (vii),

$l_{21}u_{12} + l_{22} = -4$

$2.2 + l_{22} = -4$ $[\,u_{12} = 2\,; l_{21} = 2\,]$

$4 + l_{22} = -4$

$\therefore l_{22} = -8$ -------------------------------------(xiv)

From (viii),

$$l_{21}u_{13} + l_{22}u_{23} = 6$$

$$2.3 - 8u_{23} = 6 \qquad [\, l_{21} = 2\,; u_{13} = 3\,; \therefore l_{22} = -8\,]$$

$$-8u_{23} = 6 - 6$$

$$u_{23} = 0 \qquad \text{-------------------------------------(xv)}$$

From (x),

$$l_{31}u_{12} + l_{32} = -9$$

$$3.2 + l_{32} = -9 \qquad [\, l_{31} = 3\,; u_{12} = 2\,]$$

$$l_{32} = -9 - 6$$

$$l_{32} = -15 \qquad \text{-------------------------------------(xvi)}$$

From (xi)

$$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3$$

$$3.3 - 15.0 + l_{33} = -3 \quad [\, l_{31} = 3\,; u_{13} = 3\,; l_{32} = -15\,; u_{23} = 0\,]$$

$$l_{33} = -3 - 9$$

$$l_{33} = -12 \qquad \text{-------------------------------------(xvii)}$$

$$\therefore L = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}$$

$$\Rightarrow L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix}$$

And

$$U = \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

From (i),

$$\mathbf{Au = b}$$

$$\mathbf{LUu = b} \qquad \text{[from (ii), A = LU]}$$

$$\mathbf{LUu = b} \qquad \text{-------------------------------------(xviii)}$$

Let,

$$\mathbf{Uu = v} \qquad \text{-------------------------------------(xix)}$$

Where,

Where, $\mathbf{v} = \begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \\ \mathbf{v_3} \end{pmatrix}$

From (xviii),

$$\mathbf{LUu = b}$$
$$\mathbf{Lv = b}$$

$$\Rightarrow \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{2} & \mathbf{-8} & \mathbf{0} \\ \mathbf{3} & \mathbf{-15} & \mathbf{-12} \end{pmatrix} \begin{pmatrix} \mathbf{v_1} \\ \mathbf{v_2} \\ \mathbf{v_3} \end{pmatrix} = \begin{pmatrix} \mathbf{5} \\ \mathbf{18} \\ \mathbf{6} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \mathbf{1 \times v_1 + 0 \times v_2 + 0 \times v_3} \\ \mathbf{2 \times v_1 - 8 \times v_2 + 0 \times v_3} \\ \mathbf{3 \times v_1 - 15 \times v_2 - 12 \times v_3} \end{pmatrix} = \begin{pmatrix} \mathbf{5} \\ \mathbf{18} \\ \mathbf{6} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \mathbf{v_1} \\ \mathbf{2v_1 - 8v_2} \\ \mathbf{3v_1 - 15v_2 - 12v_3} \end{pmatrix} = \begin{pmatrix} \mathbf{5} \\ \mathbf{18} \\ \mathbf{6} \end{pmatrix}$$

$$\mathbf{v_1 = 5} \qquad \text{-------------------------------------(xx)}$$
$$\mathbf{2v_1 - 8v_2 = 18} \qquad \text{-------------------------------------(xxi)}$$
$$\mathbf{3v_1 - 15v_2 - 12v_3 = 6} \qquad \text{-------------------------------------(xxii)}$$

From (xxi),

$$\mathbf{2v_1 - 8v_2 = 18}$$

$$\mathbf{2.5 - 8v_2 = 18} \qquad [\,\mathbf{v_1 = 5}\,]$$

$$\mathbf{10 - 8v_2 = 18}$$

$$\mathbf{-8v_2 = 18 - 10}$$

$-8v_2 = 8$

$\therefore v_2 = -1$ -------------------------------------(xxiii)

From (xxii),

$3v_1 - 15v_2 - 12v_3 = 6$

$3.5 - 15(-1) - 12v_3 = 6$ $[\, v_1 = 5\,;\, v_2 = -1\,]$

$15 + 15 - 12v_3 = 6$

$30 - 12v_3 = 6$

$-12v_3 = 6 - 30$

$-12v_3 = -24$

$\therefore v_3 = 2$ -------------------------------------(xxiv)

From (xix),

$Uu = v$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 \times u_1 + 2 \times u_2 + 3 \times u_3 \\ 0 \times u_1 + 1 \times u_2 + 0 \times u_3 \\ 0 \times u_1 + 0 \times u_2 + 1 \times u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_1 + 2u_2 + 3u_3 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$u_1 + 2u_2 + 3u_3 = 5$ -------------------------------------(xxv)

$u_2 = -1$ -------------------------------------(xxvi)

$u_3 = 2$ -------------------------------------(xxvii)

From (xxv),

$$\mathbf{u_1 + 2u_2 + 3u_3 = 5}$$

$$\mathbf{u_1 + 2(-1) + 3.2 = 5}$$

$$\mathbf{u_1 - 2 + 6 = 5}$$

$$\mathbf{u_1 = 5 - 4}$$

$$\mathbf{u_1 = 1}$$

$\mathbf{u_1 = 1}$, $\mathbf{u_2 = -1}$, $\mathbf{u_3 = 2}$ Answer

Example 71: Given the following system of linear equations, determine the value of each of the variables using the LU decomposition method.

$$6x_1 - 2x_2 = 14$$
$$9x_1 - x_2 + x_3 = 21$$
$$3x_1 - 7x_2 + 5x_3 = 9$$